

AWS CONCEPTUAL STUDY GUIDE

A DBA-oriented foundation: infrastructure, storage, networking, security, availability, operations, serverless, IaC, and containers.

WHO THIS IS FOR

An Oracle Apps DBA with Linux and networking familiarity who wants an AWS mental model before moving into hands-on labs.

How to use this guide

- Read in order once. The chapters build from AWS geography and identity into networking, databases, availability, operations, and automation.
- Use the boxed examples as production design patterns, not copy-and-paste configurations.
- For every workload, ask: what is the source of truth, where is state stored, how does it recover, and who is allowed to change it?

Learning sequence: Foundation -> Compute and storage -> IAM and VPC -> Databases and HA -> Operations/DR -> Modern deployment.

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TERMINOLOGY NOTE

AWS naming changes over time, but the concepts in this guide are deliberately stable. Learn the decision-making model first; then use current AWS documentation for service limits, pricing, engine support, and exact configuration.

1. AWS mental model and geography

AWS is a catalog of managed infrastructure. Instead of buying and operating every server, network, disk, database, and monitoring system in a data center, you provision services through the Console, CLI, SDKs, or infrastructure code.

AWS Account

- + - Region (for example: Mumbai / ap-south-1)
- + - Availability Zone (isolated data-center group)
 - + - VPC (your private logical network)
 - + - Subnets
 - + - EC2, RDS, load balancers, containers

Term	Meaning	DBA analogy
AWS account	Administrative, security, and billing boundary.	A distinct enterprise tenancy/cost center.
Region	Geographic AWS area containing multiple AZs.	A primary site or DR location.
Availability Zone (AZ)	Physically separate, low-latency fault domain within one Region.	Separate data-center group.
VPC	Logically isolated virtual network.	Your virtual data-center network.
Subnet	An IP range within one AZ.	A network segment/VLAN-style boundary.

DESIGN PRINCIPLE

A Region is not one building and an AZ is not a mere rack. Design across at least two AZs for production availability, then use a second Region only when DR requirements justify it.

2. EC2, EBS, AMIs, and gold images

EC2 is a virtual server. You select an AMI, instance type, VPC/subnet, security group, IAM role, and storage. **EBS** is persistent block storage attached to EC2.

Traditional concept	AWS equivalent	Important point
Linux VM	EC2 instance	Compute ends when an instance terminates.
OS/data disk	EBS volume	Persistent but normally usable only in its AZ.
Disk backup	EBS snapshot	Use for volume recovery/copy workflows.
VM template	AMI	Reusable bootable image.
Approved VM template	Gold image	A controlled, versioned AMI.

A **gold image** is a prebuilt, tested, approved AMI. It contains stable, reusable items - OS patches, hardened baseline, agents, packages, and perhaps application prerequisites. It is not a place for passwords, private keys, or environment-specific hostnames.

```
Base RHEL/Ubuntu AMI
-> patch and harden
-> install monitoring/security agents
-> test and scan
-> approve and version
-> Gold AMI: ebs-app-rhel9-2026.09.18-v3
-> launch consistent EC2 instances
```

Related term	Meaning
Image baking	Building and validating an AMI.
Immutable infrastructure	Build a new image and replace servers instead of manually editing running ones.
Configuration drift	Unexpected differences created by manual server changes.
Rollback	Redeploy the prior known-good image.

EBS LIFECYCLE

Stopping an instance stops compute charges but keeps EBS charges. Terminating an instance removes the VM; whether each EBS volume survives depends on its delete-on-termination setting. Review this setting for important data volumes.

3. Amazon S3 object storage

S3 stores objects (files plus metadata) in buckets. It is not a Linux filesystem and its folder appearance is a key-prefix convention.

```
Bucket: company-backups-prod
+- exports/prod/database-export.dmp
+- oracle/rman/2026/09/18/
+- logs/apps/2026/09/18/access.log
```

The full object name is its key. A prefix such as `oracle/rman/` groups keys.

Need	Best-fit service	Reason
Boot disk, datafile block I/O	EBS	Attached block storage.
Backups, exports, logs, artifacts	S3	Highly durable object storage.
Shared POSIX-like filesystem	EFS	Managed shared file system.
Relational application data	RDS / Oracle	Database semantics and transactions.

S3 term	Meaning
Bucket	Top-level object container; names are globally unique.
Object / key	Stored data and its full identifier.
Versioning	Retains previous object versions after overwrite/delete.
Lifecycle rule	Moves/expunges data automatically based on age.
Storage class	Cost and retrieval tradeoff - Standard, Standard-IA, Glacier tiers.
Pre-signed URL	Temporary access to a defined object without exposing credentials.

Illustrative backup lifecycle

Day 0-30 S3 Standard

Day 31-90 S3 Standard-IA

Day 91-365 Glacier Flexible Retrieval

After 365 Expire only if policy and retention requirements permit

SECURITY BASELINE

Keep Block Public Access enabled unless access is intentionally public. Use versioning for important data, encryption (often KMS for sensitive workloads), least-privilege bucket/IAM policies, and cross-Region replication when DR requires it.

4. IAM: identity and access management

IAM decides **who** is making an AWS request and **what** that identity may do. Its most important concept is the **role**: a temporary, assumable identity for workloads and people.

EC2 instance with IAM role

- > receives temporary AWS credentials automatically
- > uploads only to s3://company-prod-backups/oracle/prod/*
- > no access key stored in a script

Term	Meaning
Principal	The identity making a request: user, role, service, or account.
IAM user	Long-lived identity; generally avoid for normal workforce access.
IAM role	Assumable identity with short-lived credentials - standard for workloads.
Policy	JSON document that allows or denies actions on resources.
STS	AWS service that issues temporary credentials for assumed roles.
IAM Identity Center	Workforce SSO/federation service.

Simplified policy evaluation

Explicit deny -> DENY

Otherwise an applicable allow -> ALLOW

Otherwise -> DENY (implicit deny)

- Use the root user only for exceptional account-level tasks; enable MFA and never create root access keys.
- Use SSO/Identity Center for people and roles for EC2, Lambda, ECS, and cross-account work.
- Grant the minimum actions and resources required - not wildcard administrator permissions.
- Store passwords, private material, and API tokens in Secrets Manager or Parameter Store, never in AMIs, source control, or user-data scripts.
- Use CloudTrail and IAM Access Analyzer to investigate changes and improve permissions.

5. VPC networking

A VPC is your AWS network. It has a CIDR range, such as 10.0.0.0/16; subnets split the range and each subnet belongs to exactly one AZ.

VPC 10.0.0.0/16

AZ-A: public 10.0.1.0/24 private app 10.0.11.0/24

AZ-B: public 10.0.2.0/24 private app 10.0.12.0/24

private database subnets: separate, no public IPs

Component	What it does
Internet Gateway (IGW)	Allows direct internet connectivity for public subnet resources with public IPv4 addresses.
NAT Gateway	Allows private subnet resources to initiate outbound internet traffic without allowing unsolicited inbound traffic.
Route table	Determines packet destination; most-specific matching route wins.
Security group	Stateful resource-level firewall; allow-only rules.
Network ACL	Stateless subnet-level filter; allow/deny rules and explicit return paths.

Public subnet route: 0.0.0.0/0 -> Internet Gateway

Private subnet route: 0.0.0.0/0 -> NAT Gateway

Local VPC traffic: 10.0.0.0/16 -> local

PREFERRED DATABASE RULE

Attach app instances to app-sg and databases to db-sg. On db-sg, allow TCP 1521 from app-sg, not broadly from a CIDR such as 10.0.0.0/16. This expresses the real trust relationship.

Private connectivity	Use case
Site-to-Site VPN	Encrypted on-premises-to-VPC connection over the internet.
Direct Connect	Dedicated private circuit for sustained enterprise use.
Transit Gateway	Hub connecting many VPCs, VPNs, and Direct Connect.
VPC Peering	Simple direct VPC-to-VPC link; non-transitive.

6. RDS and Oracle on AWS

Amazon RDS is a managed relational-database service. AWS manages the underlying host and portions of patching, backups, storage, and failover. You remain responsible for data, database access, application behavior, schema, SQL, parameter choices, and recovery validation.

Area	RDS for Oracle	Self-managed Oracle on EC2
OS/root access	No	Yes
Oracle binary/host management	Controlled managed model	You manage it
Parameter flexibility	Supported parameter/option groups	Full control
Automated backups and failover	Built in options	You design and operate
EBS/legacy customization	May be constrained	Better suited when full control is required

RDS term	Meaning
DB instance	Managed database server.
DB subnet group	Approved private subnets/AZs where RDS can run.
Endpoint	Stable DNS name for client connections.
Parameter group	Managed database parameter baseline.
Multi-AZ	Synchronous standby and automated failover for HA.
Read replica	Asynchronous replication primarily for read scaling; not the same as Multi-AZ.
Snapshot / PITR	Point-in-time restore mechanisms.

Multi-AZ (availability)

Primary in AZ-A -- synchronous replication --> Standby in AZ-B

Application uses one endpoint; standby is not normally a read target.

Read replica (read scale)

Primary -- asynchronous replication --> replica

Writes go to primary; selected reads can go to replica.

ORACLE DBA CAUTION

For Oracle EBS and self-managed Oracle, validate the exact Oracle/EBS release, licensing, support matrix, feature requirements, and operational controls before selecting RDS. Do not assume managed RDS is a universal replacement for EC2-hosted Oracle.

7. Load balancing, Auto Scaling, and state

A single application EC2 instance is a single point of failure. An **Application Load Balancer (ALB)** routes HTTP/HTTPS requests to healthy targets, while an **Auto Scaling Group (ASG)** maintains the required number of EC2 instances.

Users

- > ALB across AZs
- > app instance / ECS task in AZ-A
- > app instance / ECS task in AZ-B
- > private database endpoint

Service	Role
ALB	Layer 7 HTTP/HTTPS routing; TLS termination; host/path routing; health checks.
NLB	Layer 4 TCP/UDP/TLS forwarding, often when static IPs or non-HTTP traffic is needed.
Target group	Set of back-end EC2, IP, or container targets.
ASG	Maintains min/desired/max EC2 count and replaces unhealthy instances.
Launch template	ASG definition of AMI, type, role, disks, networking, and startup behavior.

Desired capacity: 2 | Minimum: 2 | Maximum: 6

Instance fails -> ASG launches replacement

Demand rises -> scale out based on CPU, requests, or custom metrics

Stateful versus stateless is the central scaling question. A server can be safely replaced only when important state is outside that specific server and is recoverable by its replacement.

State type	Wrong local location	External durable/shared location
User session	App server RAM	ElastiCache/Redis, database, or signed token
Uploaded file	Local EBS disk	S3 or EFS
Configuration/secrets	AMI or script	Parameter Store / Secrets Manager
Logs	Local filesystem	CloudWatch Logs / S3
Application data	App server files	RDS / Oracle database

DATABASE EXCEPTION

An Oracle primary database is stateful: datafiles, redo, control files, transaction state, and database memory matter. An ALB cannot coordinate Oracle replication or transactions, and an ASG does not make a database HA. Use Oracle-aware designs such as Data Guard, RAC, RDS Multi-AZ where appropriate, plus backups and tested recovery.

8. CloudWatch and CloudTrail

Use **CloudWatch** to understand service behavior and **CloudTrail** to understand AWS API activity and changes.

Metrics tell you symptoms.

Logs explain details.

Audit trails explain changes.

Service	Question answered	Examples
CloudWatch Metrics	How is it behaving?	CPU, DB connections, EBS latency, ALB target response time.
CloudWatch Logs	What did the workload say?	Application logs, Oracle alert/listener logs, OS logs.
CloudWatch Alarms	When should we act?	RDS free storage low for 15 minutes -> SNS alert.
CloudTrail	Who changed AWS?	Security group opened, IAM policy edited, EC2 terminated.

Illustrative operational flow

RDS free storage < 15 GB for 15 minutes

-> CloudWatch alarm

-> SNS notification

-> email / Slack / incident system

-> runbook: verify growth, reclaim or extend capacity, document action

- Centralize logs before ASG/ECS replacements discard local files.
- Set retention deliberately; unlimited retention is rarely free or useful.
- Alert on actionable conditions: backup failure, database storage risk, connection failure, unhealthy targets, replication lag, and high latency.
- Use CloudTrail with an encrypted, protected S3 destination. CloudTrail is infrastructure audit; it does not replace Oracle unified auditing.
- Use AWS Config/config rules for configuration history and drift/compliance visibility where appropriate.

9. Backup, DR, and multi-Region design

High availability protects against instance or AZ failure. **Disaster recovery** protects against broader failure, including regional loss, major operational error, ransomware, or data corruption. Multi-AZ is HA; it is not automatically DR.

Term	Question	Example
RPO - Recovery Point Objective	How much data can be lost?	RPO 15 minutes: at most 15 minutes of data loss.
RTO - Recovery Time Objective	How long may recovery take?	RTO 4 hours: restore service within four hours.

DR strategy	Cost	Recovery profile	Idea
Backup and restore	Low	Highest RTO/RPO	Restore systems from backups after disaster.
Pilot light	Low-medium	Moderate	Keep critical data/replication ready; start app capacity when needed.
Warm standby	Medium-high	Low	Run a smaller usable environment in DR Region; scale during failover.
Active/active	High	Lowest	Both Regions serve traffic; highest complexity.

Primary Region	DR Region
App tier + primary Oracle DB	-- redo --> standby Oracle DB
Backups to S3	-- copy --> DR-region backup copies
Precreated VPC/IAM/DNS	+ failover runbook

- EBS snapshots are useful volume recovery points but do not by themselves prove Oracle-consistent database recovery.
- For self-managed Oracle, use RMAN, archived redo, an understood catalog/control-file strategy, and periodic isolated restore tests.
- Use S3 versioning, cross-Region replication, AWS Backup policies, and immutable/retention controls according to governance needs.
- Avoid split brain: do not allow both old primary and DR copy to accept independent writes during failover.
- Test restores. A green backup job proves data was written; it does not prove the application can be recovered inside RTO/RPO.

RECOVERY EXERCISE

Select a backup -> restore in an isolated network -> recover database -> start it -> run integrity/application checks -> measure real RTO and recovery point -> document gaps. This is the only credible validation.

10. Serverless: Lambda, API Gateway, and SQS

Serverless means AWS operates the servers. Your responsibility is code, configuration, IAM, event design, cost, limits, observability, and downstream safety. **Lambda** runs short code in response to an event.

Service	Purpose	DBA-oriented example
Lambda	Short event-driven code	Check backup status; send alert; rotate/check a secret.
EventBridge	Schedules and event routing	Run a check every 15 minutes.
API Gateway	Managed HTTP API front door	Expose approved internal automation API.
SQS	Durable asynchronous queue	Buffer report or processing requests.
SNS	Publish notifications	Notify email, Slack, or incident tooling.

EventBridge schedule (every 15 minutes)

- > Lambda checks RDS free space / backup status
- > normal: log metric
- > problem: SNS -> email / Slack / incident system

Client -> API Gateway -> Lambda -> DynamoDB/RDS/other service

Application creates report request -> SQS queue -> Lambda/worker processes it
Worker failure -> message can be retried -> repeated failures go to DLQ

Concept	Why it matters
Idempotency	Events can be delivered more than once; a retry must not duplicate a harmful action.
Timeout	Lambda has a maximum runtime; use workers/workflows for long-running jobs.
Concurrency	Limit parallel execution so a burst does not overload the database.
Cold start	New execution environments can add startup latency.
Execution role	Use an IAM role with precisely scoped permissions.
DLQ	Keeps repeatedly failed events for investigation rather than silent loss.

SCOPE RULE

Lambda is suited to lightweight, retry-safe automation. It is not generally the place for an hours-long RMAN run, an Oracle database, or a continuously running WebLogic stack.

11. Infrastructure as Code (IaC)

Infrastructure as Code defines the desired infrastructure in version-controlled files. The files are the source of truth; AWS resources are the deployed result.

```
infrastructure/  
+- network.tf  
+- security.tf  
+- compute.tf  
+- database.tf  
+- variables.tf
```

Code change -> validate -> plan/change set -> review -> deploy -> verify

Concept	Meaning
CloudFormation	AWS-native IaC using YAML/JSON templates and stacks.
Terraform	Multi-provider IaC tool using HCL and a managed state file.
Stack	CloudFormation-managed collection of deployed resources.
Change set / plan	Preview of infrastructure changes before applying them.
State file	Terraform mapping from code resources to deployed resources; treat as sensitive.
Drift	Out-of-band/manual changes causing deployed reality to differ from code.

Terraform concept

Desired state in .tf files

versus

Actual state in AWS

-> plan (proposed actions)

-> apply (perform reviewed actions)

	CloudFormation	Terraform
Ownership	AWS-native	Third-party, multi-cloud ecosystem
Language	YAML / JSON	HCL
State	Managed by CloudFormation	Team manages protected remote state
Strong fit	AWS-only/native governance	Cross-cloud or standard IaC workflow

SAFE CHANGE RULE

Never treat an IaC deploy as harmless. Read the plan/change set. A small-looking change can replace a database, alter a route, expose a firewall rule, or create downtime. Reviewers should evaluate both the intended business outcome and the exact resource actions.

12. Containers: Docker, ECS, and Fargate

A **container** packages application code, runtime, dependencies, and a start command. Unlike an EC2 VM, it does not contain a full guest OS; it shares the host kernel. Containers are usually ideal for stateless web/API/worker tiers.

VM on EC2	Container
Full guest OS	Shares host OS kernel
Often long-lived	Usually disposable/replaced
Patch server and app	Build new image and deploy replacement
Good for Oracle/legacy/custom OS needs	Good for stateless services

Dockerfile -> build -> container image -> registry -> running container

Developer builds image -> ECR repository -> ECS launches it

ECS term	Meaning
ECR	AWS registry for container images.
Task definition	Blueprint: image, CPU/memory, port, IAM, logs, network.
Task	Running copy of a task definition.
Service	Maintains desired task count and deployment behavior.
Cluster	Logical ECS grouping where tasks run.
Fargate	AWS-managed compute runtime for containers.
ECS on EC2	Containers run on EC2 instances you manage.

Users -> ALB -> ECS service
 -> Fargate task in AZ-A
 -> Fargate task in AZ-B
 -> database/private services

MENTAL MODEL

AMI : EC2 instance = container image : running container/task. In both cases, build once, test, and deploy identical versions. The container model makes replacement and scaling natural, but state still belongs in RDS, S3, EFS, ElastiCache, queues, or another appropriate durable service.

13. Reference architecture and operating checklist

Users

- > Route 53 DNS
- > CloudFront (optional cache/edge)
- > Application Load Balancer
- > Stateless EC2 ASG or ECS/Fargate tasks across AZ-A and AZ-B
- > Private RDS / self-managed Oracle database
 - > backups to S3
 - > CloudWatch metrics/logs
 - > HA/DR: Multi-AZ, Data Guard, or recovery design

IAM roles, KMS, Secrets Manager, CloudTrail, and IaC apply across the design.

Layer	Primary concern	Typical AWS choices
Entry	DNS, TLS, caching, web routing	Route 53, ACM, CloudFront, ALB
Application	Replaceable runtime and scaling	EC2 ASG or ECS/Fargate
Data	Authoritative state and recovery	RDS or Oracle on EC2; S3/EFS/ElastiCache as needed
Network	Segmentation and least access	VPC, private subnets, SGs, VPN/DX
Security	Identity, secrets, encryption, audit	IAM, KMS, Secrets Manager, CloudTrail
Operations	Observability and automation	CloudWatch, Config, EventBridge, Lambda
Repeatability	Controlled provisioning	CloudFormation or Terraform

Before approving a production design, walk through these questions:

- **State:** If each component is deleted, how does it rebuild? Where is the durable source of truth?
- **Availability:** Which AZ/component failures are tolerated? Is capacity present in another AZ?
- **Recovery:** What are the RPO/RTO? When was recovery last tested?
- **Network:** Is every database private? Are security-group rules narrowly scoped?
- **Identity:** Do workloads use roles rather than keys? Are people using SSO/MFA?
- **Observability:** Which metrics, logs, alarms, dashboards, and audit records are required?
- **Change:** Is infrastructure in code? Will a reviewed plan show the exact impact?
- **Cost:** Do availability, data transfer, storage, NAT, logs, and DR choices fit the actual business requirement?

ONE-SENTENCE SUMMARY

AWS architecture is the disciplined placement of compute, state, identity, networking, recovery, and automation into managed services - with every failure mode and permission boundary made explicit.

Appendix: key terms and your next learning path

Term	Short definition
ALB	HTTP/HTTPS load balancer with application-aware routing.
AMI	Reusable EC2 machine image.
ASG	Group that maintains/replaces/scales EC2 instances.
AZ	Isolated availability zone inside an AWS Region.
CIDR	IP address range notation, e.g. 10.0.0.0/16.
EBS	Persistent block volume for EC2.
ECS / Fargate	AWS container orchestrator / serverless container compute.
IAM role	Temporary, assumable identity and permissions.
KMS	Encryption-key management service.
NAT Gateway	Private-subnet outbound internet access.
RPO / RTO	Maximum acceptable data loss / recovery duration.
S3	Durable object storage.
Security group	Stateful, resource-level virtual firewall.
VPC	Isolated AWS virtual network.

A recommended transition from concept to practice:

Phase	Study / build objective
1. Foundation	Create a safe learning account, enforce MFA and budgets, learn Console and CLI without creating expensive resources.
2. Networking	Draw and then build a VPC with public/private subnets, IGW, NAT, and security groups.
3. Compute	Launch a hardened EC2 instance from an AMI, attach a role, and send logs to CloudWatch.
4. Storage and IAM	Use a role to write a test backup artifact to an encrypted, versioned S3 bucket.
5. HA and operations	Build a tiny ALB + ASG application pattern and simulate an instance failure.
6. IaC	Recreate the lab using CloudFormation or Terraform; inspect every plan/change set.
7. DBA specialization	Study Oracle-on-EC2 and RDS options, RMAN/S3 workflows, Data Guard, monitoring, and DR testing.

FINAL HABIT

Whenever you encounter a new AWS service, classify it: Is it compute, storage, network, identity, database, observability, security, integration, or automation? Then ask how it handles availability, permission, cost, and recovery.